

SEBI & Capital Markets

Building an Indian Regulatory Compliance AI

Group 1

Prof. Subhabrata Majumdar | Term IV, PGPBA 2026-27

Name	Roll No
2517027	Pavan Kulkarni
2517067	Badireddi Bhavya Sri
2517024	Om Tayade
2517008	Saurabh Thakre
2517059	Sahil Gupta
2517048	Agniv Ghosh



The real risk is a plausible wrong answer under a filing clock

SEBI compliance is fragmented, frequently amended, judgment-heavy and expensive to get wrong. A fluent answer with the wrong deadline is worse than no answer

Difficulty was set by construction, not intuition



50

questions across 7 rulebooks
and 5 constructed difficulty
tiers

300

scored answers

3 base models + 3 RAG variants
× 50 identical questions
× 3 independent human scorers

Rubric: factual accuracy 4 • completeness 2 • confidence
calibration 2

We tested solution against compliance questions firms face most often

The 50-question benchmark covers seven SEBI domains plus two cross-cutting provisions with the greatest weight on continuous obligations

CONTINUOUS OBLIGATIONS

22

LODR + PIT



EVENT-DRIVEN + SPECIALIST DOMAINS

16

ICDR + SAST

8 questions each

5

AIF

category-specific limits

5

MF + FPI

3 + 2 specialist
questions

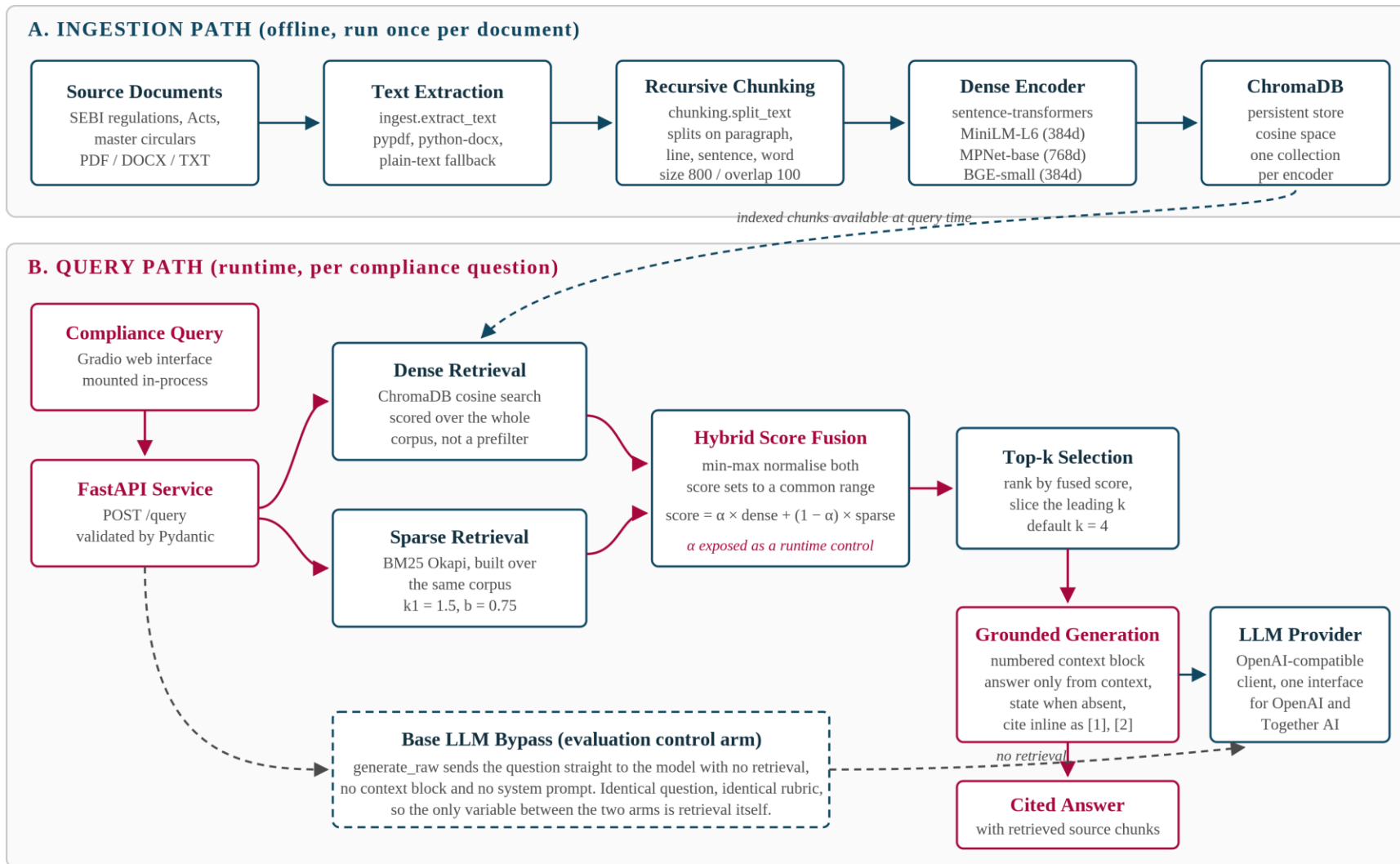
2

Cross-cutting

SEBI Act + SCRR

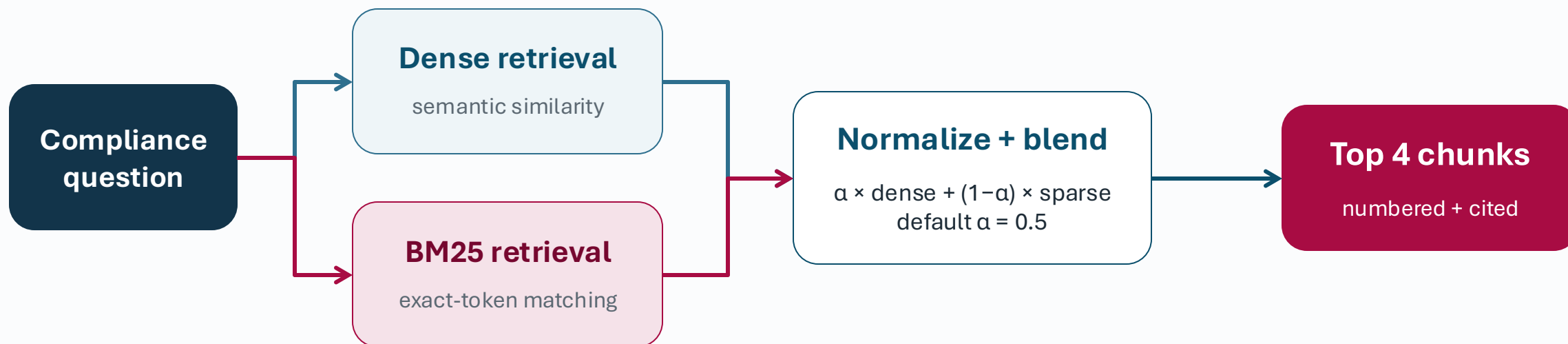
Investment signal: the benchmark combines settled provisions with recent amendments so stale-model risk becomes measurable

One pipeline turns statutory text into a source-linked answer



Hybrid retrieval protects the details legal questions hinge on

Semantic search finds paraphrases. BM25 finds the exact regulation number, threshold or deadline. We score the entire corpus with both before selecting anything



800-character chunks

100-character overlap

whole-corpus fusion

strict closed-book
prompt

Trade-off: reliable at this corpus size, production needs an inverted index, deduplication and regulation-aware chunk metadata

We built an evaluation package, not just a single RAG run

The brief set a minimum We expanded it to test whether grounding, not model size, drives the result

BRIEF MINIMUM

1

RAG configuration



OUR DELIVERED SYSTEM

3

RAG variants

same retrieval pipeline

300

scored answers

six side-by-side arms

3

encoder choices

MiniLM · MPNet · BGE

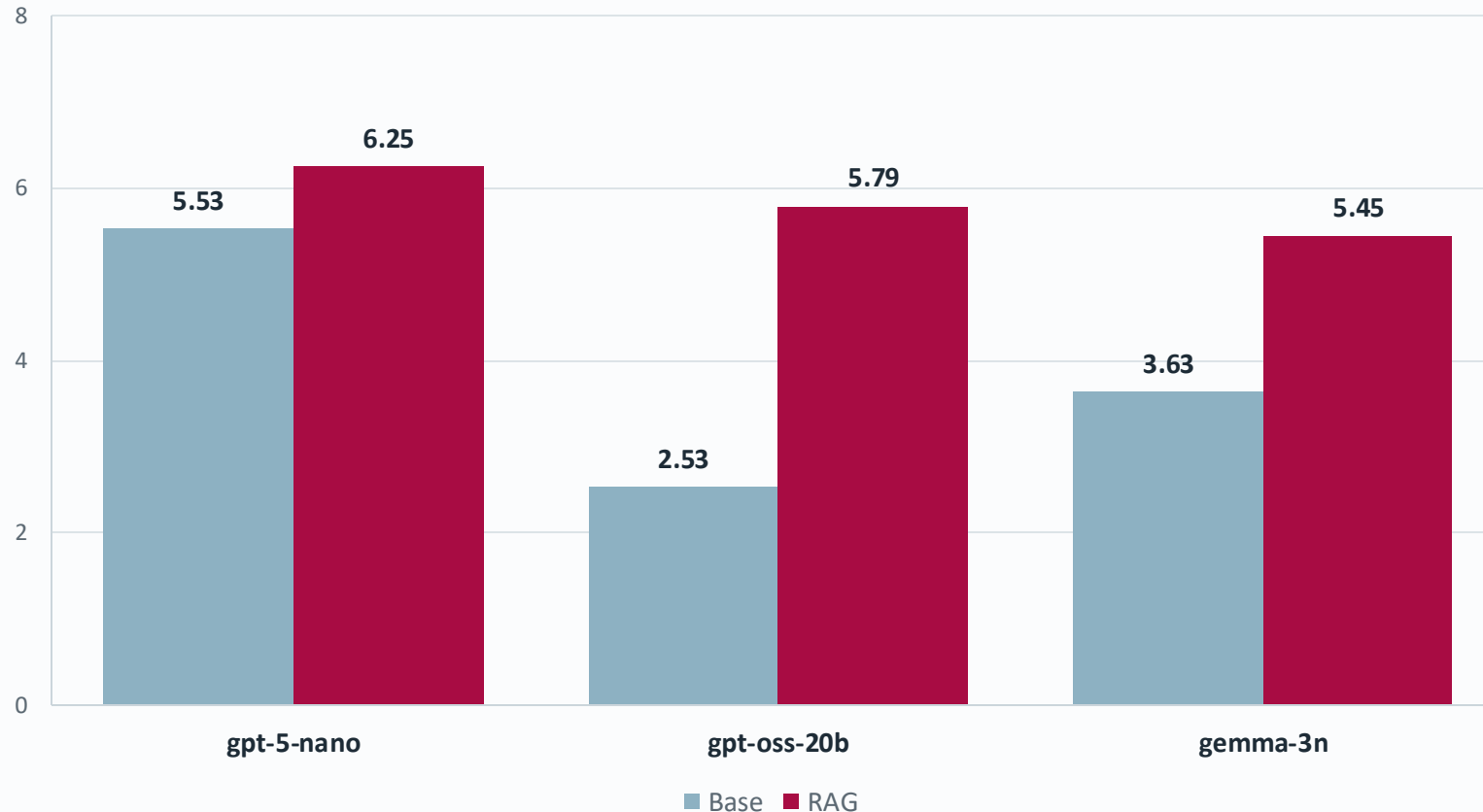
Live

operating layer

FastAPI + Gradio

Strict grounding was a deliberate control: use only retrieved context, cite it, and abstain when the answer is absent

Grounding improves every model & reduces dependence on expensive model intelligence



+1.93

average points from RAG

+3.26

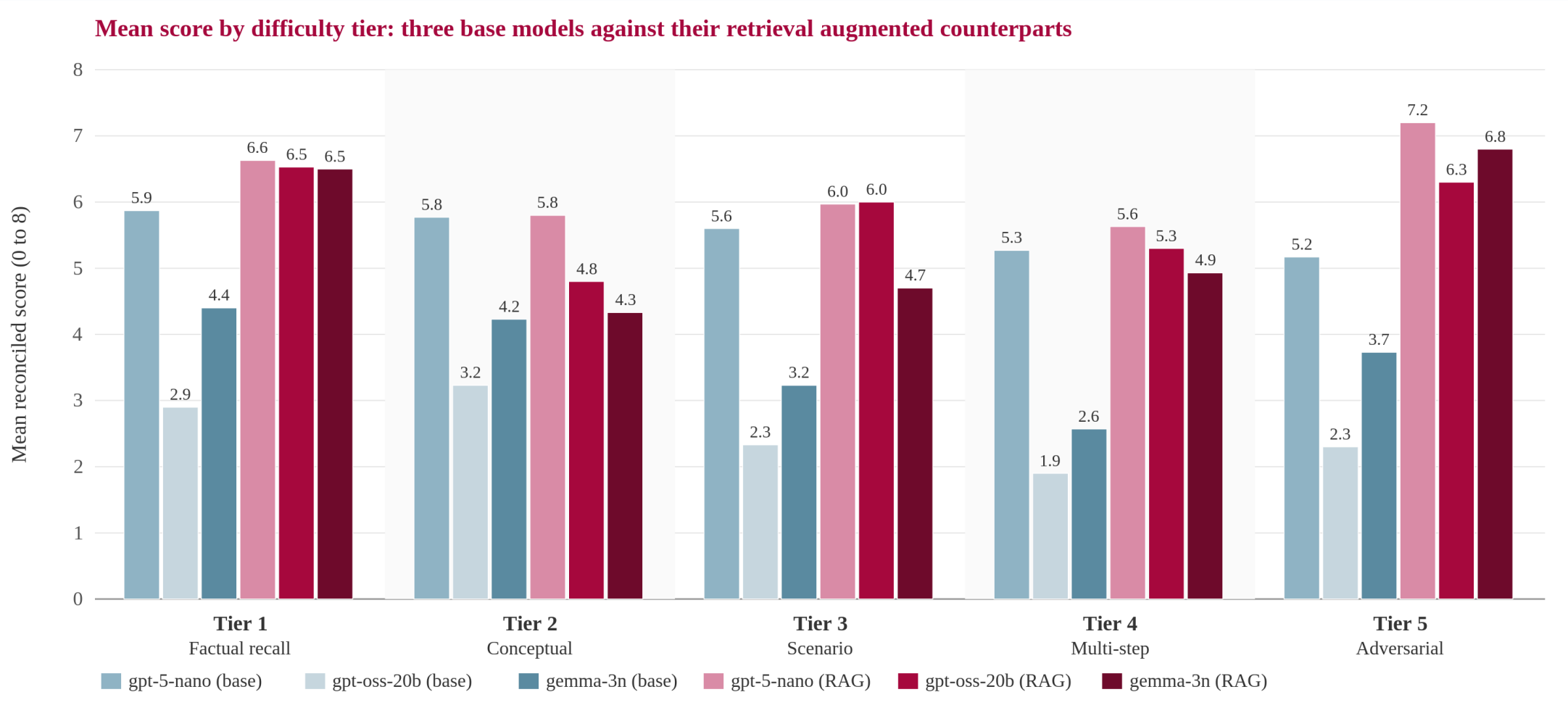
gpt-oss-20b matched-model gain

3.00 → 0.80

model performance gap

Same questions | same rubric | retrieval is the only intervention | average gain +1.93

Tier 5 is where retrieval earns its keep



Best-model uplift

Tier 5: +2.03

Tier 2: +0.03

Retrieval is targeted, not universal

Deploy RAG where it creates the highest compliance value

WHERE IT HELPS MOST



Tier 5 adversarial

fresh rules reject false premises

WHERE IT HELPS LEAST



Tier 2 conceptual

strict context can obstruct synthesis

MATCH THE CONTROL TO
THE QUESTION

Route

1.

Facts → RAG

2.

Judgment → review

Investment implication: automate source-dependent facts and retain human review for judgment

The biggest win: silent invention became visible abstention

BASE: 150 ANSWERS

61

confidently wrong

wrong and stated flatly

GROUND WITH
RETRIEVED RULES →

31 refusals

visible retrieval misses, not
hidden model guesses

RAG: 150 ANSWERS

0

confident errors

across 150 grounded answers

Business implication: Retrieval-first systems reduce compliance risk

Back a secure self-hosted compliance assistant

RECOMMENDED MODEL

RAG +
gpt-oss-20b

5.79 / 8

0.46 points below gpt-5-nano, but
deployable inside the firm

Why it wins

Keep UPSI and the vector store
inside the enterprise perimeter

AUTOMATION SCOPE

Tiers 1-2

answer with cited provisions

Tiers 3-5

draft only with compliance officer review

NON-NEGOTIABLE GUARDRAILS

- 1 Show the retrieved provision
- 2 Refuse below a similarity threshold
- 3 Pin an 'as-of' date and re-index amendments
- 4 Log every query, answer and review
- 5 Keep model + vector store on-premise

Operating principle: make verification cheaper than re-research, and
escalation cheaper than a wrong filing

What the next 90 days will deliver



90-day outcome

Validated performance | enterprise security | measurable business value

From Proven Performance to Enterprise Readiness

The proof of concept already shows measurable performance, safety and deployment advantages

TODAY

+1.93

average quality gain



THE INVESTMENT CASE

+3.26

largest model lift

gpt-oss-20b

5.79

open-model RAG
score

within 7% of best RAG

0

confident errors

across 150 grounded
answers

Scale

enterprise
platform

secure and extensible

But... should you buy?

Demo

Enterprise Guardrails and Advanced Intelligence Features

Benchmarking accuracy is only Step 1. If it cannot be trusted, it is as good as non-existent.

THE GOAL

Transforming RAG
into an Audit-Ready
Enterprise
Compliance
Platform



SELLABLE FEATURES

98.4%

Grounding Index -
Faithfulness Audit

RT NLI Sentence-level
hallucination
verification

-85%

Effort - Statutory
Timelines

Automated T + X filing
calendar & risk severity
heatmap

92%

Agreement - LLM
Jury Bench

3-model concurrent
consensus verdict for
high-stakes decisions

+18.4%

Precision Uplift -
HyDE Rewriter

Hypothetical document
embeddings boosting
Tier 5 retrieval

With your investment, we will turn proven performance into a secure, scalable enterprise platform for regulatory intelligence